



Asphalt**Mine** Database Tutorial

Version 1.0.0

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Overview

AsphaltMine Database is a platform designed for collecting asphalt test results in an organized online database. This database will be used to build machine learning models that can predict asphalt performance, significantly reducing the need for physical testing and potentially enabling the design of better-performing asphalt mixtures. The development of AsphaltMine platform is funded by the [Swiss National Science Foundation](#) (Project [No.213163](#)) as part of the "Fate of Polymers in Recycled Asphalt" project.

This tutorial will guide you through the basic steps of using AsphaltMine Database (data.asphalt-mine.org), including how to enter and explore data.

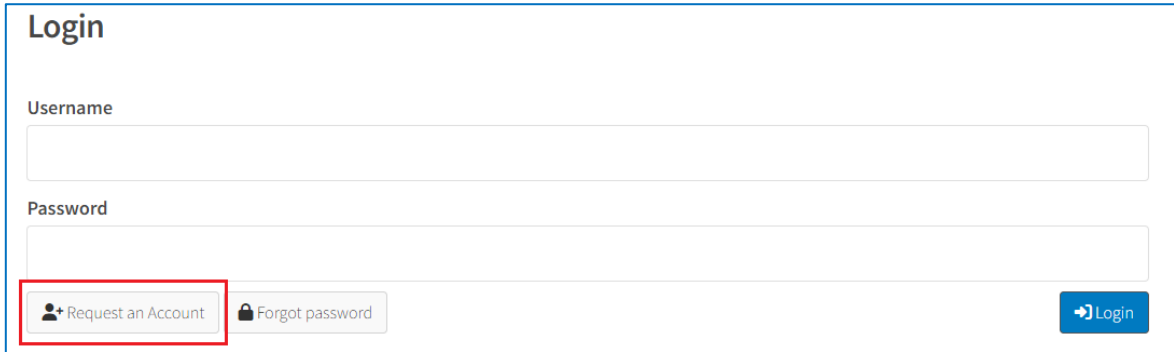
Accounts

Although the database is not publicly accessible, users can request access to the database web application through a login process. Once a request is submitted, the admin will review it and grant access. After approval, the user will be able to log in to the web application.

Gain Access

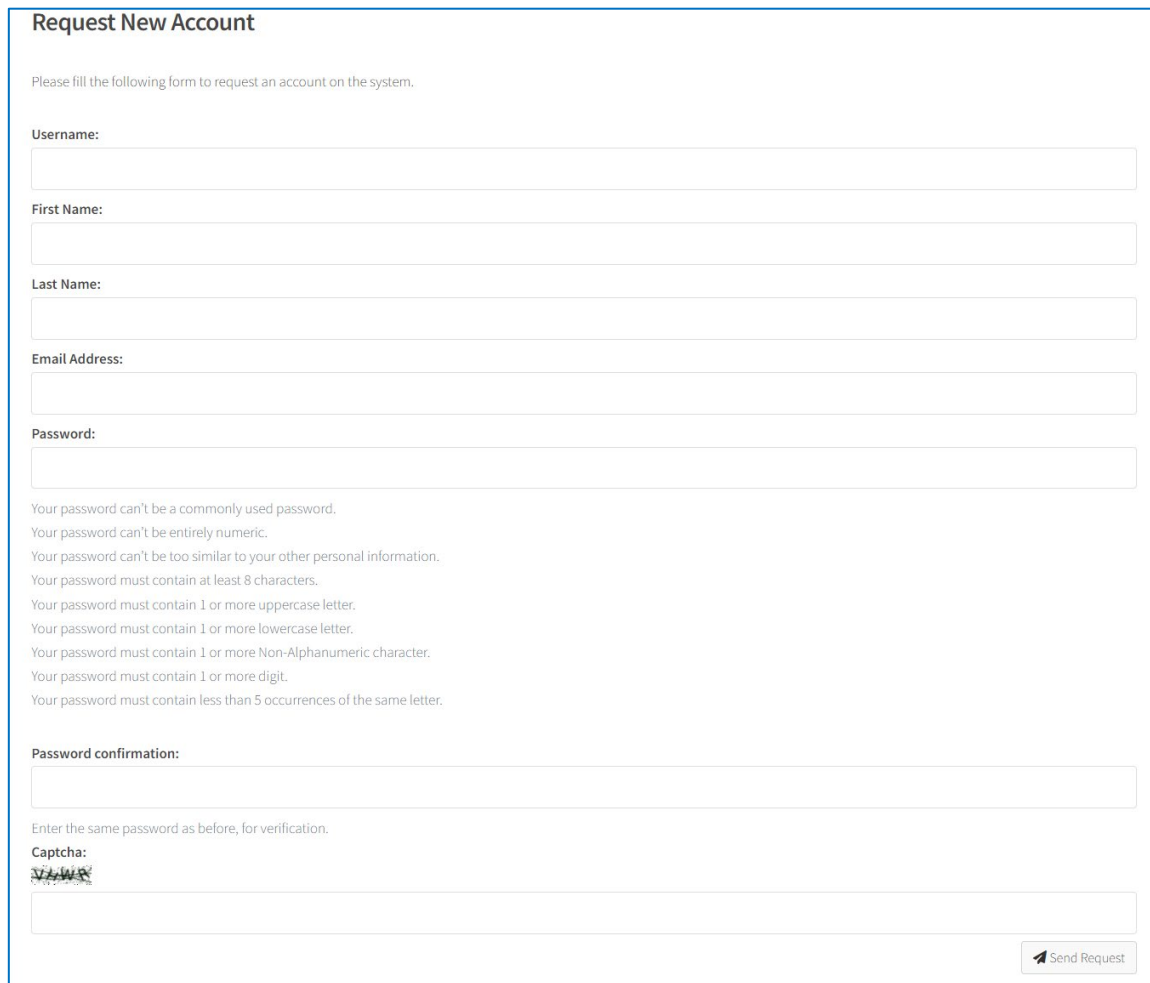
To gain access to the database, users should follow these steps:

- 1) On the login screen, click on *Request an Account* as shown below.



The screenshot shows a 'Login' form with two input fields: 'Username' and 'Password'. Below these fields are three buttons: 'Request an Account' (highlighted with a red box), 'Forgot password', and 'Login'. The 'Login' button is blue with a right-pointing arrow.

- 2) A new screen will appear as shown below. Fill in all the required fields. The **Username** is the name you will use to log in, and the **Password** is the login password.



The screenshot shows a 'Request New Account' form. It starts with a heading 'Request New Account' and a sub-heading 'Please fill the following form to request an account on the system.' Below this are several input fields: 'Username:', 'First Name:', 'Last Name:', 'Email Address:', 'Password:', and 'Password confirmation:'. Below the 'Password:' field is a list of password requirements. At the bottom, there is a 'Captcha:' field with a distorted image and a 'Send Request' button.

Username:

First Name:

Last Name:

Email Address:

Password:

Your password can't be a commonly used password.
Your password can't be entirely numeric.
Your password can't be too similar to your other personal information.
Your password must contain at least 8 characters.
Your password must contain 1 or more uppercase letter.
Your password must contain 1 or more lowercase letter.
Your password must contain 1 or more Non-Alphanumeric character.
Your password must contain 1 or more digit.
Your password must contain less than 5 occurrences of the same letter.

Password confirmation:

Enter the same password as before, for verification.

Captcha:

Send Request

- 3) Once you have completed the form, click *Send Request*. A confirmation message will appear in the top right corner of the screen, indicating that your request has been sent.

After the admin approves your request, you will be notified and can then log in using your username and password.

Access levels

Two main access levels to the database are available:

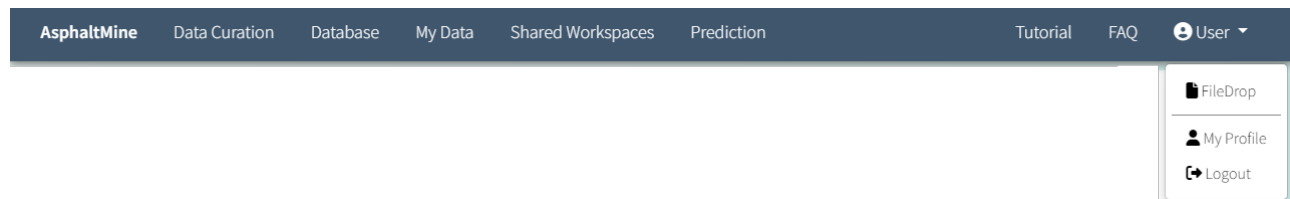
- 1) **View Records:** Users with this access level can view all open data or data to which they have been granted access by others. They can also download the data.
- 2) **Add Records:** Users with this access level can save new records to the database. Additionally, they can edit their own records in the database.

These access levels are typically granted to users together.

Navigator and Pages

Overview

The navigator contains several links to the pages within the web application, as shown below:



The pages are:

- **AsphaltMine:** This link directs the user to the homepage of asphaltmine.org.
- **Data Curation:** This directs the user to the GV Web Form.
- **Database:** This page displays a list of all records the user is authorized to view and download.
- **My Data:** This page lists the records the user owns.
- **Shared Workspaces:** This page displays all the workspaces the user is a part of, including public workspaces.
- **Prediction:** This link directs the user to the machine learning prediction tool.
- **Tutorial:** This page offers a detailed guide on the functionalities of AsphaltMine database.
- **FAQ:** This page contains a list of frequently asked questions.
- **Account:** This is a dropdown menu for user profile options, including FileDrop, logout, and, for administrators, the admin page.
- **EXA:** This page contains the EXA plugin adaption, which is used to mass import similar records (To be added later).
- **FileDrop:** This page allows users to upload files in any format, including Excel and PDF, and to attach to an existing record.
- **My Profile:** This page contains information about the user's account.

Data Curation

Overview

The Data Curation page contains the Generate-View Web Form, called the GV Form. The GV Form is a tool that allows users to add records for a selected test to the database. The form components are as follows:

- 1) **Upload Form:** Located at the top of the page, this section includes tools for uploading a record either from the database or from an external source.
- 2) **Fields Form:** This is the main form, situated just below the Upload Form. It contains all the input fields required for creating a new record, as well as for editing or viewing an existing record.
- 3) **Left Sidebar:** Used for navigating the Fields Form, this sidebar expands different sections and automatically scrolls to bring them into view.
- 4) **Right Sidebar:** This sidebar includes a tab for selecting the desired performance test and provides additional information about the form's functionality. This information can be expanded by pressing the question mark icon.

The screenshot displays the 'GV Web Form Beta' interface. On the left is a dark sidebar with navigation links: 'Data Source', 'Mixture', 'Test Results', and 'Notes'. The main area is divided into two sections. The top section, 'Upload Form', has two buttons: 'Choose Record from AsphaltMline Database' and 'Choose Record from your own Device'. Below these is a 'Select elements to upload' section with a grid of checkboxes: 'Data Source', 'Results', 'Mixture', 'General Notes', 'Sample Properties', and 'All Elements'. A 'Load' button is at the bottom of this section. The bottom section, 'Fields Form', is outlined in yellow and contains four expandable sections: 'Data Source', 'Mixture', 'Test Results', and 'Notes', each with a 'Show' button. At the bottom of this section are 'Download' and 'Save' buttons. On the right is another sidebar with a 'Performance Tests' tab showing 'Rutting Test'. Below this, it lists functionalities: 'Save new records', 'Load and View records', 'Update existing records', and 'Download records'. It also includes a note about required fields marked with an asterisk and a list of semi-mandatory fields: 'Grain Size Distribution', 'Binder Content', and 'Penetration and Voids'.

Test Selection

When selecting a performance test, the Fields Form will update immediately, adding all the fields related to the selected test and removing those associated with the unselected test. This action and its effects will be visible in both the right sidebar and the Fields Form.

For example, when the rutting test is selected, the right sidebar will display "Rutting Test" on the relevant tab, and the Fields Form will include result sections specific to this test, as shown below.

If two tests are selected, the form will display the names of both tests in the right sidebar, with each test having its own unique section contained within a new tab in the Fields Form. For example, as shown below, when both the "Rutting Test" and the "Indirect Tensile Strength Test" are selected, the form will present separate tabs for each test.

If multiple tests are selected, multiple records will be saved. The form will automatically create a record for each test by extracting the fields associated with each test from the form, building the XML document, and then sending it to the database. This process allows for the analysis of each test

individually, while also enabling the addition of data for samples with the same properties from multiple tests at the same time.

Fields Form

The Fields Form is the sole section of the GV Form, comprising various components such as outer fields, inner fields, buttons, and tooltips. The outer tabs represent the four main branches of each data record: Data Source, Mixture, Results, and Notes. Each branch is designed with a combination of these components to guide users through the data entry process.

The diagram illustrates the four main branches of the Fields Form, each represented by a tab with a 'Show' button:

- Data Source**: Contains a 'Show' button.
- Mixture**: Contains a 'Show' button.
- Test Results**: Contains a 'Show' button.
- Notes**: Contains a 'Show' button.

Within each outer tab, there are several inner tabs with different levels, as shown below. Clicking on any tab expands its corresponding section.

The diagram illustrates the nested structure of the Fields Form, showing the expansion of the 'Mixture' tab into multiple levels:

- Outer Level**: The 'Data Source' tab is highlighted with a green oval.
- First Inner Level**: The 'Mixture' tab is highlighted with a yellow oval. It contains sub-sections: 'Mixture Identifiers', 'Mixture Recipe', and 'Mixture Component Properties'.
- Second Inner Level**: The 'Mixture Component Properties' sub-section is highlighted with an orange oval. It contains 'Virgin Aggregates' and 'Filler'.
- Third Inner Level**: The 'Virgin Binder' sub-section is highlighted with a red oval. It contains a list of properties with input fields and EN standards:
 - Penetration (0.1mm): EN 1426
 - Softening Point (°C): EN 1427
 - Fraass Breaking Point (°C): EN 12593
 - Elastic Recovery (%) at 25°C: EN 13398
 - Cohesion Energy (J/cm²) at 5°C: EN 13589
 - Solubility (%): EN 12592

Below the 'Virgin Binder' section, there are additional sections: 'Viscosities' (containing 'BTSV' and 'MSCRT' with EN standards), 'Additional Properties' (containing 'Aging Resistance' with an EN standard), 'Additives', and 'Reclaimed Asphalt Aggregates'.

To take a closer look at how each section is structured, the 'Recovered Aggregates' tab is considered as an example, as shown below.

The image shows a software interface for 'Recovered Aggregates'. The main title 'Recovered Aggregates' is at the top left. A 'Collapse Tab Button' (minus sign) is at the top right. The interface contains several input fields and sections:

- Grain Size Distribution (semicolon separated)**: A section with input fields for 'Size (mm):' and 'Proportion (%)'. A 'Graph Button' (line graph icon) is next to the 'Proportion (%)' field.
- Nature:** An input field with an 'Information Icon' (i).
- Los Angeles Test Result (%):** An input field with a 'Unit' label and an 'Information Icon' (i).
- Flakiness Index (%):** An input field with an 'Information Icon' (i).
- Shape Index (%):** An input field with an 'Information Icon' (i).
- Sand Flow Coefficient (seconds):** An input field with an 'Information Icon' (i).
- Nordic Abrasion Value (%):** An input field with an 'Information Icon' (i).
- Standard Reference Number:** A list of European standards: EN 12697-2, EN 1097-2, EN 933-3, EN 933-4, EN 933-6, and EN 1097-9.
- Rounded and Crushed Proportions:** A section with an input field and a 'Standard Reference Number' label EN 933-5.
- Bitumen Coverage Degree:** A section with an input field and a 'Standard Reference Number' label EN 12697-11.
- Additional Properties:** A section with a 'Property 1' label and input fields for 'Test Method:', 'Value:', and 'Unit:'. An 'Add Property Button' (+ Property) is at the bottom.

A tooltip is visible over the 'Shape Index (%)' field, containing the text: 'The value of the sand flow coefficient of the recovered aggregate according to EN 933-6, expressed in seconds (e.g., 40).'

The components visible within each tab are as follows:

- **Main Tab Title:** The title of the main section, which is also the name of the main tab before clicking and expanding it. It represents the branch to which all fields within the section belong in the XML document.
- **Collapse Tab Button:** A button used to hide the main section. Clicking this button closes the section, making the main tab visible again.
- **Graph Button:** A button used to display a graph for specific fields, such as grain size distribution.
- **Standard Reference Number:** The European Standard number from which the field is derived. This number may appear in different locations, such as on the tab itself, near the title, or next to input elements in the form.

- **Information Icon:** An icon that shows a tooltip for each field when hovered over or clicked.
- **Tooltip:** A small pop-up explanation of the field, including an example of the expected input.
- **Field Name:** The name of the field that needs to be filled out.
- **Unit:** The unit of measurement for the field. The user must input values according to this unit.
- **Expandable Fields Tab:** A sub-tab containing additional fields that can be filled out. It functions like other tabs; once clicked, it expands, and a collapsible button appears in the top right corner.
- **Add Property Button:** A button that allows users to add additional properties to the current section. There is no limit on the number of properties that can be added.

Input Field Guidelines

Mandatory Fields

Fields marked with an asterisk (*) are required. If any required field is incomplete, the form will not be saved, downloaded, or updated. Please ensure all required fields are filled out. The form will indicate any missing fields and provide a clear message regarding the necessary information.

Semi-Mandatory Fields

Semi-mandatory fields are those where at least one must be completed. For all tests, the following fields are semi-mandatory:

- For **Grain Size Distribution**, fill at least one of the following: *Virgin Aggregates* or *Recovered Aggregates*.
- For **Binder Content**, fill at least one of the following: *Binder Content in Mixture Recipe* or *Percentage Content of Recovered Binder*.
- For **Penetration**, fill at least one of the following: *Virgin Binder Penetration* or *Recovered Binder Penetration*.
- For **Voids**, fill at least one of the following: *Bulk density with Maximum density* or *Air voids*.

Field Input Formats

Each field only accepts a specific format as input. Although users are expected to follow the standard and fill in the fields based on the appropriate units, the form will check if the input format is valid to prevent incorrect values from being entered into the database. For example, entering a percentage above 100% will raise an error and prevent the form from being submitted. The acceptable format for each field could be one of the following:

- **Number:** This can be any positive or negative number with decimal points. An example is the Softening Point Increase in Aging Resistance.
- **Positive Number:** This can be any positive number with decimal points. Examples include Temperature and Sample Thickness.

- **Positive Integer:** This can only be a positive number without decimal points. An example for this field is the number of cycles in certain tests.
- **Percentage:** This is typically any number between 0 and 100, with decimal points. An example is Binder Content. In specific cases where a negative percentage is allowed, this will be indicated in the tooltip. In such cases, the range is between -100 and 100, including decimal points. An example of negative percentages is the Mass Loss in Aging Resistance.
- **Degree Angle:** This can be any number between 0 and 90, with decimal points. An example is the BTSV Phase Angle.
- **Year:** This field can only contain a year.
- **String:** This field can accept any characters, words, sentences, or groups of sentences.

Error Messages

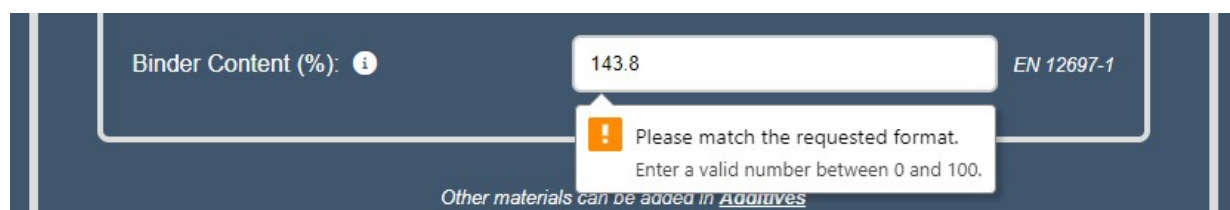
Several error messages can occur when an incorrect field is filled in, highlighting the error and scrolling it into view. Here is a list of the most common errors:

- **Empty field:** This error occurs if a required field is left empty. A default browser error message will appear on the screen, as shown below.



The screenshot shows a form with two fields: "Measurement Campaign ID: * ⓘ" and "Organization Name: * ⓘ". The "Organization Name" field is empty, and a tooltip with an orange exclamation mark icon and the text "Please fill out this field." is displayed over it.

- **Incorrect Format:** This error occurs when the value entered is in the wrong format. A default browser error will appear on the screen, along with a message stating the expected format, as shown below.



The screenshot shows a form with a field labeled "Binder Content (%): ⓘ" containing the value "143.8". To the right of the field is the text "EN 12697-1". A tooltip with an orange exclamation mark icon and the text "Please match the requested format. Enter a valid number between 0 and 100." is displayed over the field. Below the field, there is a link: "Other materials can be added in [Additives](#)".

- **Incorrect Entry Errors:** This error can occur in graph fields, such as grain size distributions. If the two graph fields do not have the same number of values, an error will appear, clearly stating the names of the two fields on the screen, as shown below.

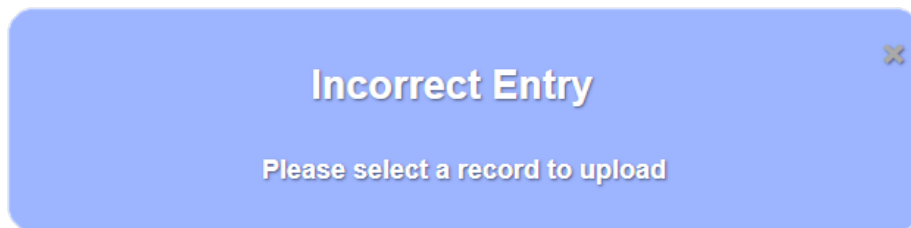
Incorrect Entry ✕

Aggregates Distribution mismatch: The number of Size values and Proportion values must be the same.

Additionally, if one of the two fields contains at least one incorrect format, a message will appear, clearly stating the field name and the incorrect format, as shown below.



A case occurs when the "Load" button in the upload form is clicked without selecting any records using the 'Choose Record' buttons. In this case, an error message should clearly indicate that a record is expected, as shown below.

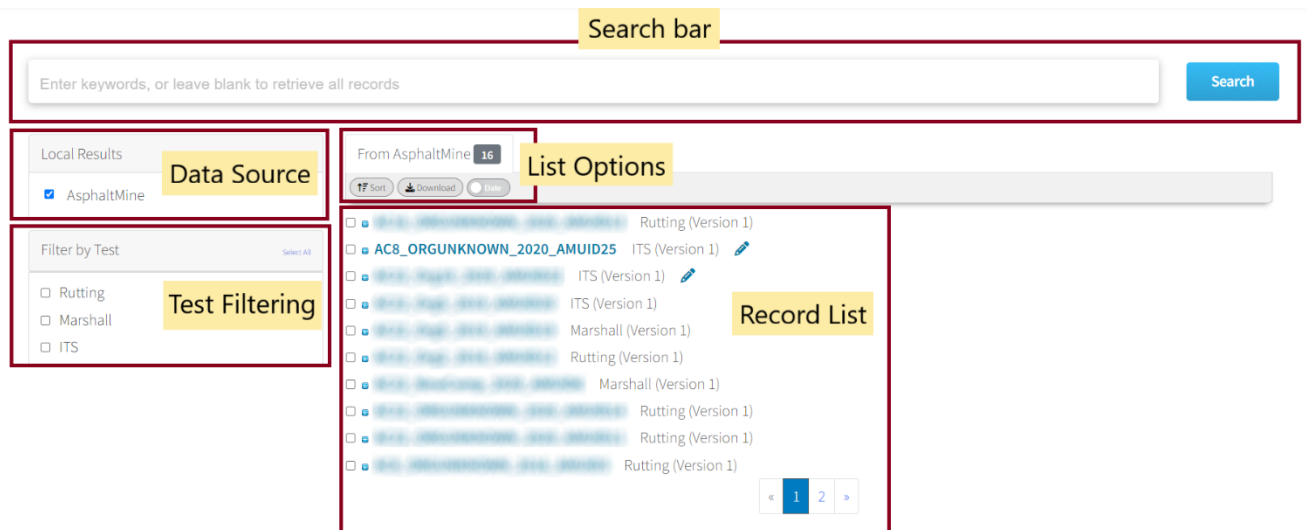


In general, the form checks the values entered in each field and validates them. If a value is incorrect, an error message will appear for that specific field before moving to the next one. Thus, errors will not be displayed all at once but will be presented one by one, from the first field to the last.

It should also be noted that in all fields expecting a number with a decimal point, the limit is four digits after the decimal point. This requirement is not explicitly stated anywhere, and a format error only appears if the limit is breached. It is important to use no more than four digits after the decimal point, although the standards typically require only one or two digits for the measured properties.

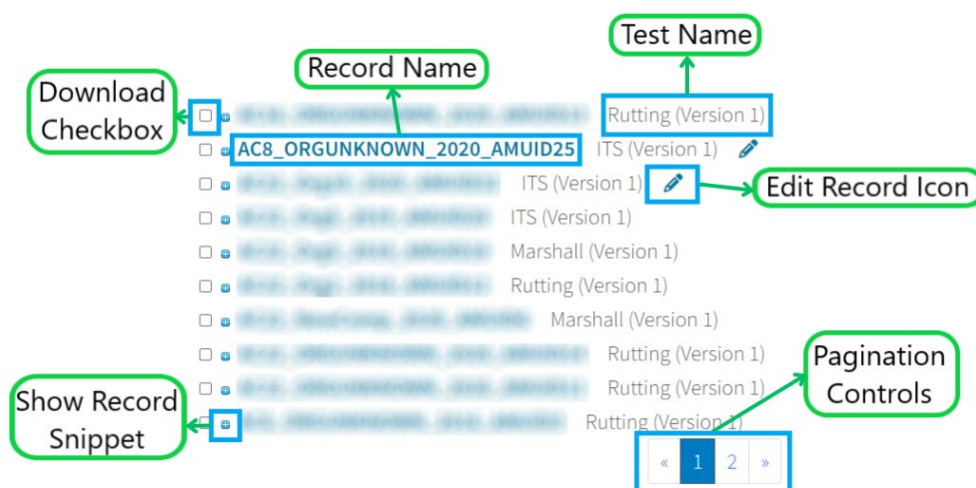
Database

This page contains the list of all records that the user is authorized to view, edit and download. The page can be divided into several sections as shown below.

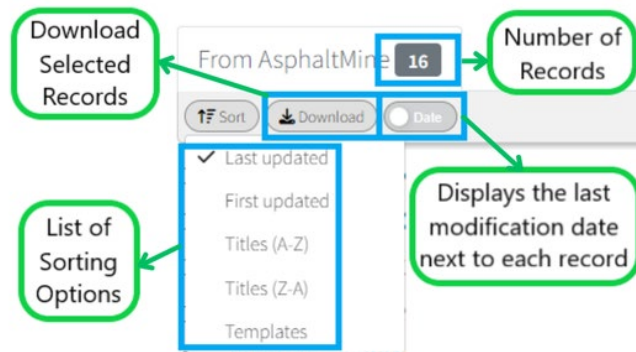


The sections are:

- **Search bar:** This enables the user to search for specific information within the records. Once the search is executed, records containing the searched information are listed, while others are excluded.
- **Data Source:** This shows the source of the records, which is AsphaltMine. The checkbox for this source is ticked by default and must always remain ticked.
- **Test Filtering:** This allows the user to filter records by specific tests. If no tests are selected, all records will be shown.
- **Record List:** This section is the core part of the database. It lists the records and allows the user to navigate through them.



- **List Options:** This section contains tools that can affect the Record List, including sorting options, a download button, and a date toggle button, as shown below. It also displays the total number of records shown in the Record List, which may vary depending on the account and selected exploration conditions.



My Data

This page displays all the records the user owns, as shown below.

Total number of records: 2

Name	Test	Last Modification date	Workspace	Actions
AC8_ORGUNKNOWN_2020_AMUID25	Rutting (Version 1)	2020-10-20 10:00:00	None	
AC8_ORGUNKNOWN_2020_AMUID25	ITS (Version 1)	2020-10-20 10:00:00	Global Public Workspace	

 View

 Edit

 Change workspace

 Change Owner

 Delete

The information contained in this page includes:

- **Total Number of records:** Displays the total number of records on this page, representing all records the user owns.
- **Name:** Shows the name of the record.
- **Test:** Indicates the test type associated with the record.
- **Last Modification date:** Displays the date of the most recent modification to the record.
- **Workspace:** Shows the workspace the record belongs to. If none, the record has not yet been assigned to a workspace.
- **Actions:** This button opens a dropdown menu with the following options:
 - **View:** Opens the record in the GV Form for viewing.
 - **Edit:** Opens the record in edit mode within the GV Form.
 - **Change Workspace:** Allows the user to assign the record to a different workspace.
 - **Change Owner:** Allows changing the record's owner, transferring rights such as deleting the record.
 - **Delete:** Deletes the record.

Shared Workspaces

This page displays all the workspaces the user is a part of, including public workspaces.

My workspaces

Total number of workspaces: 3

Create workspace

Title	Owner	Can read	Can write	Public	Actions
Global Public Workspace	GLOBAL	True	False	True	View Content
trial	UserTrial	True	False	False	View Content
trial	UserHL	True	True	False	View Content Manage Access Set public Delete

The information contained in this page includes:

- **Total Number of workspaces:** Displays the total number of workspaces listed on this page, representing all the workspaces the user is a part of.
- **Owner:** Shows the username of the workspace owner.
- **Can Read:** Indicates whether the user can view the records within this workspace.
- **Can Write:** Indicates whether the user can add new records to this workspace.
- **Public:** Indicates whether the workspace is public and visible to all users.
- **Actions:** Contains several buttons with the following options:
 - **View Content:** Allows the user to view the records available in the workspace and perform actions such as viewing records, editing records, changing the workspace or owner, and deleting records, as shown below.

Global Public Workspace

Data (8) Files (0) Previous page

Name	Test	Last Modification date	User	Actions
AC22_ORGUNKNOWN_2020_AMUID14	Rutting (Version 1)	Aug 15, 2020, 1:00 PM	ADMIN	View Content
AC22_ORGUNKNOWN_2020_AMUID14	ITS (Version 1)	Aug 15, 2020, 1:00 PM	ADMIN	View Content
AC22_trial_2020_AMUID14	Rutting (Version 1)	Aug 15, 2020, 1:00 PM	ADMIN	View Content
AC22_trial_2020_AMUID14	Marshall (Version 1)	Aug 15, 2020, 1:00 PM	ADMIN	View
AC22_ORGUNKNOWN_2020_AMUID14	Rutting (Version 1)	Aug 15, 2020, 1:00 PM	ADMIN	Edit
AC22_ORGUNKNOWN_2020_AMUID14	Marshall (Version 1)	Aug 15, 2020, 1:00 PM	ADMIN	Change workspace
AC22_ORGUNKNOWN_2020_AMUID14	Marshall (Version 1)	Aug 15, 2020, 1:00 PM	ADMIN	Change Owner
AC22_ORGUNKNOWN_2020_AMUID14	ITS (Version 1)	Aug 15, 2020, 1:00 PM	ADMIN	Delete

- **Manage Access:** Allows the user to grant other users access to either read, write, or both in this workspace. In this section, the user can view all the individuals who have been granted permission to read (i.e., load and view records) or write (i.e., save and edit records) in this workspace. It is also possible to grant access to an entire "group" of users at once, provided the group has been predefined by a staff member. Only the workspace owner is allowed to view this section, shown below, and make changes within it.

[Add users](#) [Add groups](#) [← Previous page](#)

Users:

Username	Can read	Can write	Actions
user1@org	☒	☐	Remove user access
user2	☒	☒	Remove user access

Groups:

No groups have access to **Project**

The owner can add users or groups by clicking the corresponding buttons, after which a popup message will appear, as shown below. The owner will then be able to add other users to this workspace and assign them either read or write access, as preferred. Note that this popup will display the usernames of all users in AsphaltMine, except for those who have already been granted access.

Give user rights to workspace

Select all users you want to give rights to the workspace:

☐ admin ☐ user1@org

☐ Read

☐ Write

✕ Cancel

✓ Add

- **Set Public:** This button makes the workspace public, allowing it to be viewed by all users. If the workspace is already public, the owner can set it to private in the same way, where a "Set Private" button will appear instead of this button.
- **Delete:** Permanently deletes the workspace.
- **Create Workspace:** This button enables the creation of a new workspace. A popup window will prompt the user to enter the name of the workspace, as shown below. Once created, the workspace will be listed, and the owner will have the ability to manage it.

Create new workspace

Please enter the name of the new workspace.

Workspace name:

✕ Cancel

➕ Create Workspace

My Profile

This page contains the information about the user, as shown below.

My Profile

Preferences Edit Profile Change Password	
Name:	
Username:	User
Email:	
Admin:	False
Active:	True
Superuser:	False
Last Login:	Sept 16, 2024 1:28:45 PM
Date Joined:	Sept 16, 2024 1:28:45 PM

The information available on this page is:

- **Name:** The full name of the user. It is mandatory when creating an account.
- **Username:** The name used by the user to log in and is visible to other accounts within the web application. It is mandatory when creating an account.
- **Email:** The user's email address. This is required when setting up an account.
- **Admin:** Indicates whether the user has administrative access within the account.
- **Active:** Specifies if the account is active. Users with inactive accounts cannot log in until they contact an admin.
- **Superuser:** Indicates whether the user is the head of administrators.
- **Last Login:** Displays the date and time of the user's last login.
- **Date Joined:** Shows the date the user account was created.
- **Edit Profile:** This button opens the "Edit Profile" page, shown below, which allows the user to update their first name, last name, and email address.

My Profile - Edit

Use the following form to edit your information.

First Name:

This field is required.

Last Name:

This field is required.

Email Address:

This field is required.

✕ Cancel

Save changes

- **Preferences:** This button opens the "Preferences" page, shown below, where the user can set their account's time zone according to their preferences.

Time zone:

- **Change Password:** This button opens the "Change Password" page, shown below, enabling the user to update their account password.

Home > Password change

Password change

Please enter your old password, for security's sake, and then enter your new password twice so we can verify you typed it in correctly.

Old password:	
New password:	
	Your password can't be a commonly used password. Your password can't be entirely numeric. Your password can't be too similar to your other personal information. Your password must contain at least 8 characters. Your password must contain 1 or more uppercase letter. Your password must contain 1 or more lowercase letter. Your password must contain 1 or more Non-Alphanumeric character. Your password must contain 1 or more digit. Your password must contain less than 5 occurrences of the same letter.
New password confirmation:	

Curation of Data

Overview

Data curation can be performed by users either through the GV Form or using EXA plugin. Both options allow users to save new records to the database.

In brief, the GV Form is primarily used to add individual cases and provides a user-friendly interface for loading, viewing, and editing records. On the other hand, EXA is designed for bulk adding of records, specifically when they share the same Excel format.

Curation using GV Web Form

Curation Process

The user can save new records to the database using the GV Form. Follow these steps:

- 1) On the right sidebar, click on the performance tests tab.

The screenshot displays the GV Web Form Beta interface. On the left is a dark sidebar with navigation links: Data Source, Mixture, Test Results, and Notes. The main content area is divided into two primary sections. The top section, titled 'Upload Form', contains two buttons: 'Choose Record from AsphaltMine Database' and 'Choose Record from your own Device'. Below these is a 'Select elements to upload' section with a grid of checkboxes for Data Source, Mixture, Sample Properties, Results, General Notes, and All Elements. A 'Load' button is positioned below the grid. The bottom section contains four expandable panels labeled 'Data Source', 'Mixture', 'Test Results', and 'Notes', each with a 'Show' button. At the bottom of this section are 'Download' and 'Save' buttons. On the right sidebar, the 'Performance Tests' tab is selected and highlighted with a red box, showing a 'Rutting Test' button. Below this, a text box lists functionalities: Save new records, Load and View records, Update existing records, and Download records. A note at the bottom of the sidebar states that fields marked with an asterisk are required and lists semi-mandatory fields: Grain Size, Distribution, Binder Content, Penetration and Voids.

- 2) Select the test or tests you desire. You can select multiple tests to fill out at the same time.

GV Web Form Beta

Data Source
Mixture
Test Results
Notes

Upload Form

Choose Record from AsphaltMine Database
Choose Record from your own Device

Select Performance Tests

☒ Rutting Test
☐ Marshall Test
☐ Indirect Tensile Strength Test

Apply

Data Source
Show

Mixture
Show

Test Results
Show

Notes
Show

Download Save

Performance Tests
Rutting Test

This web form offers the following database functionalities:

- Save new records
- Load and View records
- Update existing records
- Download records

Fields marked with an asterisk (*) are required.

The following fields are semi-mandatory: Grain Size Distribution, Binder Content, Penetration and Voids.

- 3) Click Apply. The form is now ready to be filled out.
- 4) Fill in all the fields with the data you have, especially the required fields.
- 5) Click the Save button.

GV Web Form Beta

Data Source
Mixture
Test Results
Notes

Bulk Density (Mg/m³): 2.419
Method of Measurement: Bulk density by dimensions

Voids EN 12697-8

Air Voids (%): 5.6
Mineral Aggregate Voids (%):
Voids Filled With Bitumen (%):

Rut Depth versus Cycles (semicolon separated)

Cycles: 50, 250, 500, 1000, 1500, 2000, 250
Rut Depth (mm): 0.48, 0.81, 0.95, 1.11, 1.21, 1.27, 1.32, 1

+ Replication

Test Results Notes

Notes
Show

Download Save

Performance Tests
Rutting Test

This web form offers the following database functionalities:

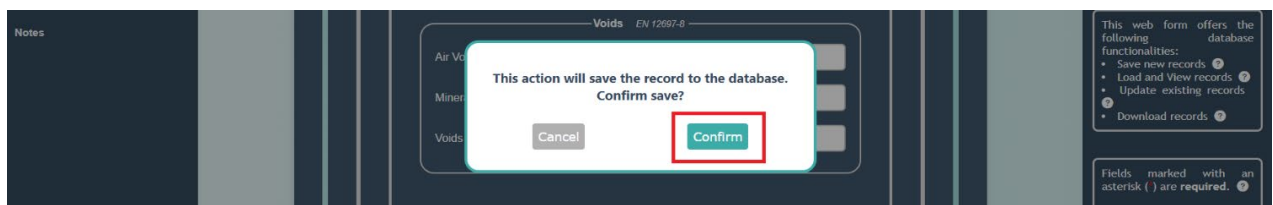
- Save new records
- Load and View records
- Update existing records
- Download records

Fields marked with an asterisk (*) are required.

The following fields are semi-mandatory: Grain Size Distribution, Binder Content, Penetration and Voids.

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- 6) Click Confirm.



A popup message confirming the data has been saved should appear in the top right corner.



Make sure to save the records to the database before exiting the page to avoid losing any data.

Stored Data Records

The saved records are stored in the MongoDB database in an XML (eXtensible Markup Language) document format. XML is a flexible, text-based format that is widely used for storing and transporting data. It allows the data to be structured hierarchically, making it easy to organize, access, and share information across different systems. By using MongoDB, the XML data is stored efficiently in a NoSQL database, which allows for flexible schema design and scalability. The use of XML ensures that the data is both human-readable and machine-readable, providing a standardized format that can be easily integrated with other applications and platforms.

Here is a sample of the Mixture branch of a record:

```
<Mixture><MixtureIdentifiers><MixtureID>PmB 45/80-65</MixtureID><MixtureType>AC 22</MixtureType></MixtureIdentifiers><MixtureRecipe><CompositionBinder><TargetBinderGrade>45/80-65</TargetBinderGrade><BinderContent>3.8</BinderContent></Binder></Composition><MixtureMaximumDensity>2.562</MixtureMaximumDensity></MixtureRecipe><Mixing><MixingMethod>Laboratory mechanical mixing</MixingMethod></Mixing></Mixture>
```

If rearranged with spaces for the different levels, the hierarchical structure of XML will be clearly visible:


```
<Mixture>
  <MixtureIdentifiers>
    <MixtureID>PmB 45/80-65</MixtureID>
    <MixtureType>AC 22</MixtureType>
  </MixtureIdentifiers>
  <MixtureRecipe>
    <Composition>
      <Binder>
        <TargetBinderGrade>45/80-65</TargetBinderGrade>
        <BinderContent>3.8</BinderContent>
      </Binder>
    </Composition>
    <MixtureMaximumDensity>2.562</MixtureMaximumDensity>
  </MixtureRecipe>
  <Mixing>
    <MixingMethod>Laboratory mechanical mixing</MixingMethod>
  </Mixing>
</Mixture>
```

Curation using EXA

EXA is a plugin developed by Beat Münch at the Concrete and Asphalt Laboratory in Empa, Switzerland (contact: beat.muench@empa.ch) for converting XML data files compatible with XSD (XML Schema Definition) into Excel data files, and vice versa.

The plugin can be downloaded from [here](#).

This page and its functionality will be made available later.

Exploration of Data

Overview

Users can explore data in the AsphaltMine database by loading, viewing, editing, deleting, and downloading records as needed. Data can be accessed either directly from the AsphaltMine database or by uploading files from a user's device. Records can be explored through various interfaces, such as the GV Form, Database, My Data, and Shared Workspaces pages.

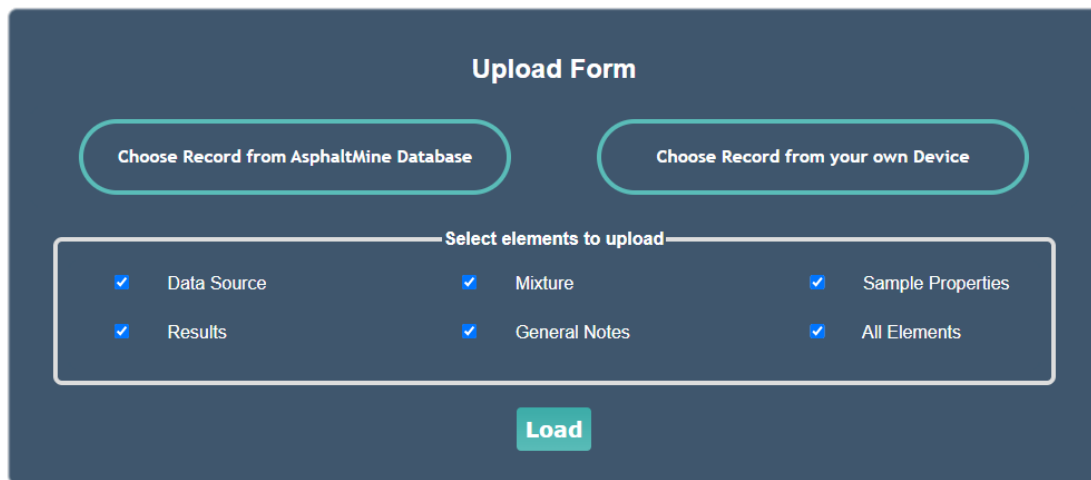
Core functionalities include:

- **Loading and Viewing Data:** Retrieve records from the AsphaltMine database or upload files from your device, and view the detailed information and graphs for each record.
- **Editing Data:** Update records to ensure data is current.
- **Deleting Records:** Remove records as needed.
- **Downloading Data:** Export records to your device.

Load and View Record

Overview

Users can load and view records through the Upload Form located at the top of the GV Form page, as shown below.



The screenshot shows the 'Upload Form' interface. At the top, there are two buttons: 'Choose Record from AsphaltMine Database' and 'Choose Record from your own Device'. Below these buttons is a section titled 'Select elements to upload' which contains a grid of six checkboxes, all of which are checked: 'Data Source', 'Results', 'Mixture', 'General Notes', 'Sample Properties', and 'All Elements'. At the bottom of the form is a green 'Load' button.

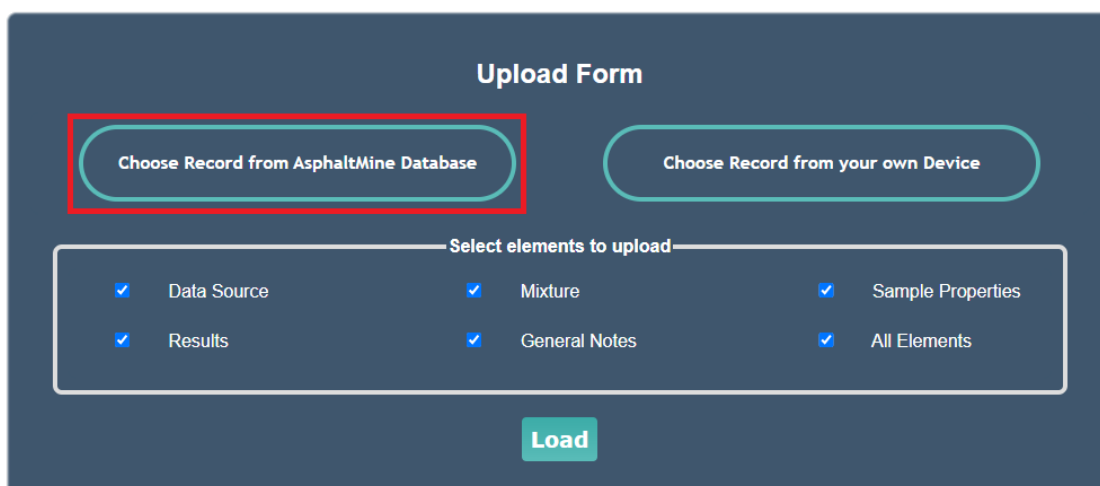
Through this form, records can be loaded either from the AsphaltMine database or from the user's own device, depending on their preference. For external uploads, the record should be in an XML format that is compatible with the tree structure of the form.

In addition to using the Upload Form, users can also load and view records directly from the database by visiting the "Database" page, "My Data" page or the "Shared Workspaces" page.

Upload from AsphaltMine database using GV Form

To upload from the AsphaltMine database using the GV Form, follow these steps:

- 1) Click the "Choose Record from AsphaltMine Database" button. This will direct you to the Database page.



This screenshot is identical to the one above, but the 'Choose Record from AsphaltMine Database' button is highlighted with a red rectangular border to indicate it is the first step in the process.

- 2) Select a record from the database by clicking on it.

Enter keywords, or leave blank to retrieve all records Search

Local Results
☒ AsphaltMine

Filter by Test
☐ Rutting
☐ Marshall
☐ ITS

From AsphaltMine 24
Sort Download Date

☐	AC16_ORGUNKNOW_2020_AMUID4	Rutting (Version 1)	
☐	AC16_ORGUNKNOW_2020_AMUID4	Rutting (Version 1)	
☐	AC16_ORGUNKNOW_2020_AMUID4	Marshall (Version 1)	
☒	AC22_ORGUNKNOW_2020_AMUID14	Rutting (Version 1)	
☐	AC16_ORGUNKNOW_2020_AMUID4	Rutting (Version 1)	
☐	AC16_ORGUNKNOW_2020_AMUID4	Rutting (Version 1)	
☐	AC16_ORGUNKNOW_2020_AMUID4	ITS (Version 1)	
☐	AC16_ORGUNKNOW_2020_AMUID4	Marshall (Version 1)	
☐	AC16_ORGUNKNOW_2020_AMUID4	ITS (Version 1)	

1 2 3

- 3) Specify the elements to upload from the list: Data Source, Mixture, Sample Properties, Results, or General Notes. By default, all elements are selected. Any elements the user unselects will not be included in the form after loading.

Select elements to upload

☒ Data Source
☐ Results

☒ Mixture
☐ General Notes

☐ Sample Properties
☐ All Elements

- 4) Click the Load button.

Upload Form

AC16_ORGUNKNOW_2020_AMUID4
Choose Record from your own Device

Select elements to upload

☒ Data Source
☐ Results

☒ Mixture
☐ General Notes

☐ Sample Properties
☐ All Elements

Load

Once clicked, the fields in the form will be populated. You can then navigate through the record using the tabs.

Upload from your own device using GV Form

To upload a record from your own database using the GV Form, follow these steps:

- 1) Click the "Choose Record from your own Device" button.

Upload Form

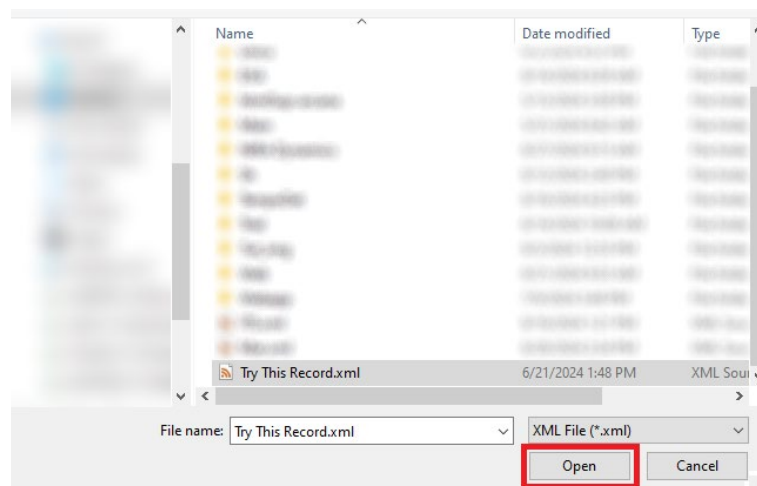
Choose Record from AsphaltMine Database
Choose Record from your own Device

Select elements to upload

☒ Data Source	☒ Mixture	☒ Sample Properties
☒ Results	☒ General Notes	☒ All Elements

Load

2) Select a record from your device and upload it.



3) Specify the elements to upload from the list: Data Source, Mixture, Sample Properties, Results, or General Notes. By default, all elements are selected. Any elements the user unselects will not be included in the form after loading.

Select elements to upload

☒ Data Source	☒ Mixture	☐ Sample Properties
☐ Results	☐ General Notes	☐ All Elements

4) Click the Load button.

Upload Form

Choose Record from AsphaltMine Database
Try This Record.xml

Select elements to upload

☒ Data Source	☒ Mixture	☒ Sample Properties
☒ Results	☒ General Notes	☒ All Elements

Load

Once clicked, the fields in the form will be populated. You can then navigate through the record using the tabs.

Upload from AsphaltMine database using "Database" page

To upload from the AsphaltMine database via the "Database" page, follow these steps:

- 1) Navigate to the "Database" page.

Enter keywords, or leave blank to retrieve all records

Local Results

☒ AsphaltMine

Filter by Test [Select All](#)

☐ Rutting

☐ Marshall

☐ ITS

From AsphaltMine 24

Sort Download Date

☐ [ID] Rutting (Version 1) [Edit]

☐ [ID] Rutting (Version 1) [Edit]

☐ [ID] Marshall (Version 1) [Edit]

☒ AC22_ORGUNKNOWN_2020_AMUID14 Rutting (Version 1) [Edit]

☐ [ID] Rutting (Version 1) [Edit]

☐ [ID] Rutting (Version 1) [Edit]

☐ [ID] Rutting (Version 1) [Edit]

☐ [ID] Rutting (Version 1) [Edit]

☐ [ID] ITS (Version 1) [Edit]

☐ [ID] Marshall (Version 1) [Edit]

☐ [ID] ITS (Version 1) [Edit]

« 1 2 3 »

- 2) Select a record to view by clicking on it. This will load the record into the GV Form.

Enter keywords, or leave blank to retrieve all records

Local Results

☒ AsphaltMine

Filter by Test [Select All](#)

☐ Rutting

☐ Marshall

☐ ITS

From AsphaltMine 24

Sort Download Date

☐ [ID] Rutting (Version 1) [Edit]

☐ [ID] Rutting (Version 1) [Edit]

☐ [ID] Marshall (Version 1) [Edit]

☒ AC22_ORGUNKNOWN_2020_AMUID14 Rutting (Version 1) [Edit]

☐ [ID] Rutting (Version 1) [Edit]

☐ [ID] Rutting (Version 1) [Edit]

☐ [ID] Rutting (Version 1) [Edit]

☐ [ID] Rutting (Version 1) [Edit]

☐ [ID] ITS (Version 1) [Edit]

☐ [ID] Marshall (Version 1) [Edit]

☐ [ID] ITS (Version 1) [Edit]

« 1 2 3 »

- 3) Specify the elements to upload from the list: Data Source, Mixture, Sample Properties, Results, or General Notes. By default, all elements are selected. Any elements the user unselects will not be included in the form after loading.

Select elements to upload

☒ Data Source ☒ Mixture ☐ Sample Properties

☐ Results ☐ General Notes ☐ All Elements

- 4) Click the Load button.

Once clicked, the fields in the form will be populated. You can then navigate through the record using the tabs.

Upload from AsphaltMine database using "My Data" page

To upload from the AsphaltMine database via the "My Data" page, follow these steps:

- 1) Navigate to the "My Data" page.

Total number of records: 19

Name	Test	Last Modification date	Workspace	Actions
AC16_ORGUNKOWN_2020_AMUID4	ITS (Version 1)	2020-08-10 10:00:00	Test	
AC16_ORGUNKOWN_2020_AMUID4	Rutting (Version 1)	2020-08-10 10:00:00	Test	
AC16_ORGUNKOWN_2020_AMUID4	Marshall (Version 1)	2020-08-10 10:00:00	Test	
AC16_ORGUNKOWN_2020_AMUID4	Rutting (Version 1)	2020-08-10 10:00:00	Test	
AC16_ORGUNKOWN_2020_AMUID4	ITS (Version 1)	2020-08-10 10:00:00	Test	
AC16_ORGUNKOWN_2020_AMUID4	Rutting (Version 1)	2020-08-10 10:00:00	Test	
AC16_ORGUNKOWN_2020_AMUID4	Rutting (Version 1)	2020-08-10 10:00:00	Test	
AC16_ORGUNKOWN_2020_AMUID4	Rutting (Version 1)	2020-08-10 10:00:00	Test	
AC16_ORGUNKOWN_2020_AMUID4	Marshall (Version 1)	2020-08-10 10:00:00	Test	
AC22_ORGUNKOWN_2020_AMUID14	Rutting (Version 1)	2020-08-10 10:00:00	Test	

« 1 2 »

- 2) Find the record that you want to view. Unlike the Database page, this page shows only the records that you own.
- 3) For the desired record, click on the Actions button in the last column, then click view. This will load the record into the GV Form.

Total number of records: 19

Name	Test	Last Modification date	Workspace	Actions
AC16_ORGUNKNOWN_2020_AMUID4	ITS (Version 1)	2020-10-20 10:00:00	ITS	
AC16_ORGUNKNOWN_2020_AMUID4	Rutting (Version 1)	2020-10-20 10:00:00	ITS	
AC16_ORGUNKNOWN_2020_AMUID4	Marshall (Version 1)	2020-10-20 10:00:00	ITS	
AC16_ORGUNKNOWN_2020_AMUID4	Rutting (Version 1)	2020-10-20 10:00:00	ITS	
AC16_ORGUNKNOWN_2020_AMUID4	ITS (Version 1)	2020-10-20 10:00:00	ITS	
AC16_ORGUNKNOWN_2020_AMUID4	Rutting (Version 1)	2020-10-20 10:00:00	ITS	
AC16_ORGUNKNOWN_2020_AMUID4	Rutting (Version 1)	2020-10-20 10:00:00	ITS	
AC16_ORGUNKNOWN_2020_AMUID4	Rutting (Version 1)	2020-10-20 10:00:00	ITS	
AC16_ORGUNKNOWN_2020_AMUID4	Marshall (Version 1)	2020-10-20 10:00:00	ITS	
AC22_ORGUNKNOWN_2020_AMUID14	Rutting (Version 1)	2020-10-20 10:00:00	ITS	

- 4) Specify the elements to upload from the list: Data Source, Mixture, Sample Properties, Results, or General Notes. By default, all elements are selected. Any elements the user unselects will not be included in the form after loading.

Select elements to upload

☒ Data Source

☒ Mixture

☐ Sample Properties

☐ Results

☐ General Notes

☐ All Elements

- 5) Click the Load button.

Upload Form

AC16_ORGUNKNOWN_2020_AMUID4

Choose Record from your own Device

Select elements to upload

☒ Data Source

☒ Mixture

☐ Sample Properties

☐ Results

☐ General Notes

☐ All Elements

Load

Once clicked, the fields in the form will be populated. You can then navigate through the record using the tabs.

Upload from AsphaltMine database using "Shared Workspaces" page

For this method, the record must be in one of the workspaces.

To upload from the AsphaltMine database via the "Shared Workspaces" page, follow these steps:

1) Navigate to the "Shared Workspaces" page.

Total number of workspaces: 5 Create workspace

Title	Owner	Can read	Can write	Public	Actions
Global Public Workspace	GLOBAL	True	True	True	View Content
My workspace	user1	True	True	False	View Content Manage Access Set public Delete
My workspace	user2	True	True	False	View Content
My workspace	user3	True	True	False	View Content
My workspace	user4	True	True	False	View Content

2) Choose the target workspace by clicking on the "View Content" Button.

Total number of workspaces: 5 Create workspace

Title	Owner	Can read	Can write	Public	Actions
Global Public Workspace	GLOBAL	True	True	True	View Content
My workspace	user1	True	True	False	View Content Manage Access Set public Delete
My workspace	user2	True	True	False	View Content
My workspace	user3	True	True	False	View Content
My workspace	user4	True	True	False	View Content

3) Within the selected workspace, select the desired record and click on the Actions button in the last column, then click "View." This will load the record into the GV Form.

Global Public Workspace

Data (12) Files (0) Previous page

Name	Test	Last Modification date	User	Actions
AC22_ORGUNKNOWN_2020_AMUID14	Rutting (Version 1)	2020-10-10 10:10:10	user1	View
AC22_ORGUNKNOWN_2020_AMUID14	Rutting (Version 1)	2020-10-10 10:10:10	user1	View
AC22_ORGUNKNOWN_2020_AMUID14	Marshall (Version 1)	2020-10-10 10:10:10	user1	View
AC22_ORGUNKNOWN_2020_AMUID14	Rutting (Version 1)	2020-10-10 10:10:10	user1	View
AC22_ORGUNKNOWN_2020_AMUID14	Rutting (Version 1)	2020-10-10 10:10:10	user1	View
AC22_ORGUNKNOWN_2020_AMUID14	Rutting (Version 1)	2020-10-10 10:10:10	user1	View
AC22_ORGUNKNOWN_2020_AMUID14	ITS (Version 1)	2020-10-10 10:10:10	user1	View
AC22_ORGUNKNOWN_2020_AMUID14	Marshall (Version 1)	2020-10-10 10:10:10	user1	View
AC22_ORGUNKNOWN_2020_AMUID14	ITS (Version 1)	2020-10-10 10:10:10	user1	View
AC22_ORGUNKNOWN_2020_AMUID14	Marshall (Version 1)	2020-10-10 10:10:10	user1	View

« 1 2 »

4) Specify the elements to upload from the list: Data Source, Mixture, Sample Properties, Results, or General Notes. By default, all elements are selected. Any elements the user unselects will not be included in the form after loading.

Select elements to upload

☒ Data Source

☒ Mixture

☐ Sample Properties

☐ Results

☐ General Notes

☐ All Elements

5) Click the Load button.

Upload Form

AC16_ORGUNKNOWN_2020_AMUID4

Choose Record from your own Device

Select elements to upload

☒ Data Source

☒ Mixture

☐ Sample Properties

☐ Results

☐ General Notes

☐ All Elements

Load

Once clicked, the fields in the form will be populated. You can then navigate through the record using the tabs.

View Graph Fields

In graph fields, you can visualize the data by clicking the graph icon within the form, as shown below.

Rut Depth versus Cycles *(semicolon separated)*

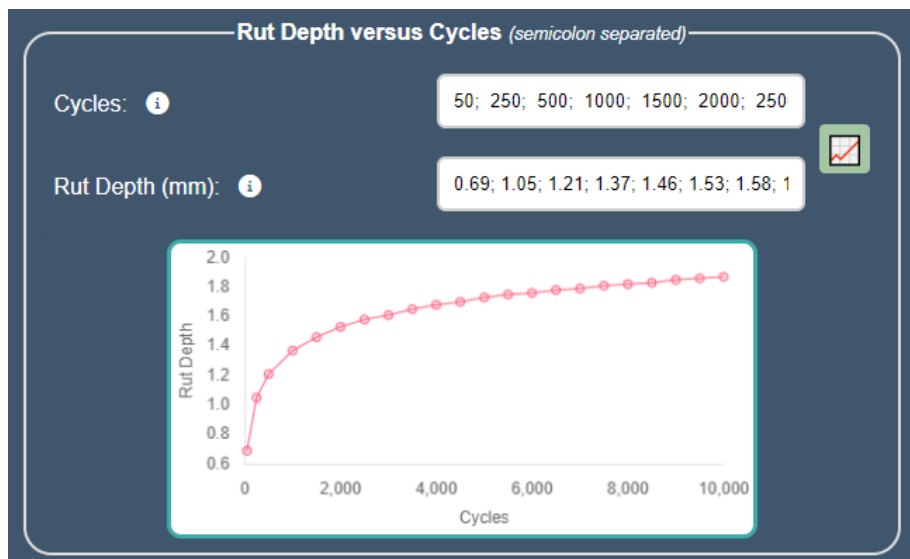
Cycles: ⓘ

50; 250; 500; 1000; 1500; 2000; 2500

Rut Depth (mm): ⓘ

0.9; 1.3; 1.47; 1.62; 1.71; 1.78; 1.83; 1.87

Once clicked, the graph will appear directly beneath the fields in their section, as shown.



The names of the axes are clear, and all input points are marked with circles on the curve. By clicking the button again, the graph will be hidden; clicking the button once more will make the graph visible again. If the user changes any values in the graph fields, the graph will not update instantly. The user must first hide the graph and then show it again by clicking the button twice. In case of incorrect value formats or if the number of values in the fields is not equal, any entry errors will appear when the button is clicked.

Edit Record

Overview

Users can update existing records directly in the database. This feature allows them to modify or correct data as needed, ensuring the database remains up-to-date with the latest information.

To edit a record, users need to open it in edit mode. This can be done through the "Database" page, the "My Data" page, or the "Shared Workspaces" page.

Editing through "Database" page

To update a record through the "Database" page, follow these steps:

- 1) Navigate to the "Database" Page.

The screenshot shows the 'Database' page interface. At the top is a search bar with the placeholder text 'Enter keywords, or leave blank to retrieve all records' and a blue 'Search' button. Below the search bar, on the left, is a 'Local Results' section with a checkbox for 'AsphaltMine' which is checked. Below that is a 'Filter by Test' section with a 'Select All' link and three checkboxes: 'Rutting', 'Marshall', and 'ITS'. On the right, there is a header 'From AsphaltMine' with a count of '16'. Below this header are buttons for 'Sort' (with a dropdown arrow), 'Download', and 'Date'. A list of records follows, each with a checkbox, a blue link to the record, and a label like 'Rutting (Version 1)' or 'ITS (Version 1)'. Some records have a blue pencil icon to their right, indicating they are editable. At the bottom of the list is a pagination bar with '<' and '>' arrows and page numbers '1' and '2'.

- 2) Identify the record to be edited. Records permitted for editing are indicated by a pencil icon.
- 3) Click on the pencil icon.

This is a close-up screenshot of the record list from the previous image. It shows the same header and buttons. The record 'AC8_ORGUNKNOWN_2020_AMUID25 ITS (Version 1)' is highlighted. A red square box is drawn around the blue pencil icon to the right of this record, indicating that this is the record to be edited.

This will open the GV Form in edit mode as shown below, and a green popup message will appear in the top right corner.

GV Web Form Beta

Data Source

Measurement Campaign ID: AC8

Organization Name: ORGUNKOWN

Location Country: SOMEWHERE

Year: 2020

Data Record Type: Select

Notes

Data Source Notes: no

Mixture

Mixture Identifiers

Mixture ID: PmB 4500-65

Mixture Type: AC 22

Mixture Recipe

Composition

Aggregates Grain Size Distribution (permeation separated) EN 833-1

Size (mm): 0.063; 0.09; 0.25; 0.75; 2; 4; 8; 11.2; 16; 2

Editing Record: "AC8_ORGUNKOWN_2020_AMUID25" "Indirect Tensile Strength Test"

This web form offers the following functionalities:

- Save new records
- Load and View records
- Update existing records
- Download records

Fields marked with an asterisk (*) are required.

The following fields are semi-mandatory: Grain Size Distribution, Binder Content, Penetration and Voids.

- 4) Modify entries, delete data, or add new information as needed.
- 5) Click the "Update" button.

GV Web Form Beta

Air Voids (%): 99

Mineral Aggregate Voids (%): 8

Voids Filled With Bitumen (%): 0.8

+ Replication

Notes

ITS Test Results Notes: 44.5

Notes

Show

Update

Editing Record: "AC8_ORGUNKOWN_2020_AMUID25" "Indirect Tensile Strength Test"

This web form offers the following functionalities:

- Save new records
- Load and View records
- Update existing records
- Download records

Fields marked with an asterisk (*) are required.

The following fields are semi-mandatory: Grain Size Distribution, Binder Content, Penetration and Voids.

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- 6) Click "Confirm".

This action will update the record in the database.
The action is irreversible. Confirm update?

Cancel Confirm

Editing through "My Data" page

To update a record through the "My Data" page, follow these steps:

- 1) Navigate to the "My Data" page.

Total number of records: 3

Name	Test	Last Modification date	Workspace	Actions
AC8_ORGUNKNOWN_2020_AMUID25	Rutting (Version 1)	Aug 11, 2020 10:00:00	Public	
AC8_ORGUNKNOWN_2020_AMUID25	ITS (Version 1)	Aug 11, 2020 10:00:00	Global Public Workspace	
AC8_ORGUNKNOWN_2020_AMUID25	Marshall (Version 1)	Aug 11, 2020 10:00:00	Global Public Workspace	

- 2) Identify the record to be edited. All records displayed on this page are available for modification.
- 3) For the desired record, click on the Actions button in the last column, then click edit. This will open the record in edit mode in the GV Form.

Total number of records: 3

Name	Test	Last Modification date	Workspace	Actions
AC8_ORGUNKNOWN_2020_AMUID25	Rutting (Version 1)	Aug 11, 2020 10:00:00	Public	
AC8_ORGUNKNOWN_2020_AMUID25	ITS (Version 1)	Aug 11, 2020 10:00:00	Global Public Workspace	
AC8_ORGUNKNOWN_2020_AMUID25	Marshall (Version 1)	Aug 11, 2020 10:00:00	Global Public Workspace	

View

Edit

Change workspace

Change Owner

Delete

This will open the GV Form in edit mode as shown below, and a green popup message will appear in the top right corner.

- 4) Modify entries, delete data, or add new information as needed.
- 5) Click the "Update" button.

The screenshot shows the 'GV Web Form Beta' interface. On the left is a sidebar with 'Data Source', 'Mixture', 'Test Results', and 'Notes'. The main area contains a form for 'Indirect Tensile Strength Test' with fields for 'Air Voids (%)', 'Mineral Aggregate Voids (%)', and 'Voids Filled With Bitumen (%)'. Below these are 'Notes' sections. A red box highlights the 'Update' button at the bottom of the form. On the right, there is a green header 'Editing Record: "ACB_ORGUNKNOWN_2020_AMUID25"', a 'Cancel' button, and a list of functionalities: 'Save new records', 'Load and View records', 'Update existing records', and 'Download records'. Below this, it states 'Fields marked with an asterisk (*) are required.' and lists semi-mandatory fields: 'Grain Size Distribution', 'Binder Content', 'Penetration and Voids'.

- 6) Click "Confirm".

The screenshot shows a confirmation dialog box with the text: 'This action will update the record in the database. The action is irreversible. Confirm update?'. There are two buttons: 'Cancel' and 'Confirm'. The 'Confirm' button is highlighted with a red box.

Editing through the "Shared Workspaces" page

For this method, the record must be in one of the workspaces.

To update a record through the "Shared Workspaces" page, follow these steps:

- 1) Navigate to the "Shared Workspaces" page.

Total number of workspaces: 3 Create workspace

Title	Owner	Can read	Can write	Public	Actions
Global Public Workspace	GLOBAL	True	False	True	<button>View Content</button>
Highway	Highway	True	False	False	<button>View Content</button>
Highway	Highway	True	True	False	<button>View Content</button> <button>Manage Access</button> <button>Set public</button> <button>Delete</button>

- 2) Choose the target workspace by clicking on the "View Content" Button.

- 4) Modify entries, delete data, and add new information as needed.
- 5) Click the Update button.

The screenshot shows the 'GV Web Form Beta' interface. On the left is a sidebar with 'Data Source', 'Mixture', 'Test Results', and 'Notes'. The main area contains a form for editing a record. At the top, a green banner reads 'Editing Record: "AC8_ORGUNKNOWN_2020_AMUID25" "Indirect Tensile Strength Test"' with a 'Cancel' button. Below this are input fields for 'Air Voids (%)', 'Mineral Aggregate Voids (%)', and 'Voids Filled With Bitumen (%)'. A '+ Replication' button is below these. A 'Notes' section contains a text area with 'ITS Test Results Notes' and the value '44.5'. Below the notes is a 'Show' button. At the bottom of the form, the 'Update' button is highlighted with a red box. On the right, a sidebar lists functionalities: 'Save new records', 'Load and View records', 'Update existing records', and 'Download records'. It also includes a note about mandatory fields marked with an asterisk.

- 6) Click Confirm.

The screenshot shows a confirmation dialog box in the center of the screen. The dialog box contains the text: 'This action will update the record in the database. The action is irreversible. Confirm update?'. Below the text are two buttons: 'Cancel' and 'Confirm'. The 'Confirm' button is highlighted with a red box. The background shows the same form as the previous screenshot, but it is dimmed.

Notes

In edit mode, the green popup message shows the names of the record and the test, and allows exiting the edit mode without saving by clicking "Cancel", as shown below.

The screenshot shows a green rectangular popup message. It contains the text: 'Editing Record: "AC8_ORGUNKNOWN_2020_AMUID25" "Indirect Tensile Strength Test"'. Below this text is a 'Cancel' button.

This popup message will not disappear, and as long as it is on the screen, it indicates edit mode. Additionally, one can notice that in this mode there is no upload form.

Another important point is that the user cannot change the type of test for the selected record in edit mode. If the user needs to change the type of test for a certain record for a certain reason, you will need to upload the record first, then save it as a new record with the different test performance type.

Download Records

Overview

Users can download selected records to their device using either the GV Form or directly through the "Database" page. The format of the selected records could either be XML or JSON.

Download using GV Form

This method allows users to download a certain record in XML format using the GV form. To do so, follow these steps:

- 1) Load an existing record using one of the methods previously described, or fill out a new one directly in the GV Form.
- 2) Click the "Download" button.

The screenshot displays the 'GV Web Form Beta' interface. On the left is a sidebar with navigation links: 'Data Source', 'Mixture', 'Test Results', and 'Notes'. The main area contains several input fields for test data: 'Diameter (mm)' with value 9, 'Bulk Density (Mg/m³)' with value 888, 'Method of Measurement' set to 'Bulk density - sealed specimen', 'Air Voids (%)' with value 88, 'Mineral Aggregate Voids (%)' with value 8, and 'Voids Filled With Bitumen (%)' with value 0.8. Below these is a '+ Replication' button. A 'Notes' section contains a text area with 'ITS Test Results Notes' and value 44.5, and a 'Show' button. At the bottom, the 'Download' button is highlighted with a red rectangle, next to a 'Save' button. The footer shows '© 2024 AsphaltMine' and 'All rights reserved'. On the right, there are two informational boxes: 'Performance Tests' with a button for 'Indirect Tensile Strength Test', and a list of functionalities (Save new records, Load and View records, Update existing records, Download records) and a note about semi-mandatory fields (Grain Size Distribution, Binder Content, Penetration and Voids).

- 3) Click "Confirm".

This screenshot shows a confirmation dialog box overlaid on the GV Web Form. The dialog box contains the text: 'This action will download the record to your device in XML format. Confirm download?'. It has two buttons: 'Cancel' and 'Confirm'. The 'Confirm' button is highlighted with a red rectangle. The background shows the same form as the previous screenshot, but it is dimmed.

If all required fields are filled, the record will download immediately. If any fields are missing, error messages will appear on the screen indicating what needs to be corrected. After fixing all errors, click the "Download" button again, then click "Confirm."

If multiple tests are selected in the Test Performance tab and completed, multiple records will be downloaded simultaneously, with one file for each test.

Download using "Database" page

This method allows users to download one or multiple records either in XML or JSON formats directly from the "Database" page. To do so, follow these steps:

- 1) Navigate to the "Database" page.

Enter keywords, or leave blank to retrieve all records Search

Local Results
☒ AsphaltMine

Filter by Test Select All
☐ Rutting
☐ Marshall
☐ ITS

From AsphaltMine 25
tf Sort Download Date

☐	ITS (Version 1)	
☐	Rutting (Version 1)	
☒	AC8_ORGUNKNOWN_2020_AMUID25 ITS (Version 1)	
☐	ITS (Version 1)	
☐	ITS (Version 1)	
☐	Rutting (Version 1)	
☐	ITS (Version 1)	
☐	Marshall (Version 1)	
☐	Marshall (Version 1)	
☐	Rutting (Version 1)	

« 1 2 3 »

- 2) Select the record or records to download by checking the box next to each record.

Enter keywords, or leave blank to retrieve all records Search

Local Results
☒ AsphaltMine

Filter by Test Select All
☐ Rutting
☐ Marshall
☐ ITS

From AsphaltMine 25
tf Sort Download Date

☐	ITS (Version 1)	
☐	Rutting (Version 1)	
☒	AC8_ORGUNKNOWN_2020_AMUID25 ITS (Version 1)	
☒	ITS (Version 1)	
☒	ITS (Version 1)	
☐	Rutting (Version 1)	
☐	ITS (Version 1)	
☒	Marshall (Version 1)	
☒	Marshall (Version 1)	
☐	Rutting (Version 1)	

« 1 2 3 »

- 3) Click "Download".

Enter keywords, or leave blank to retrieve all records Search

Local Results
☒ AsphaltMine

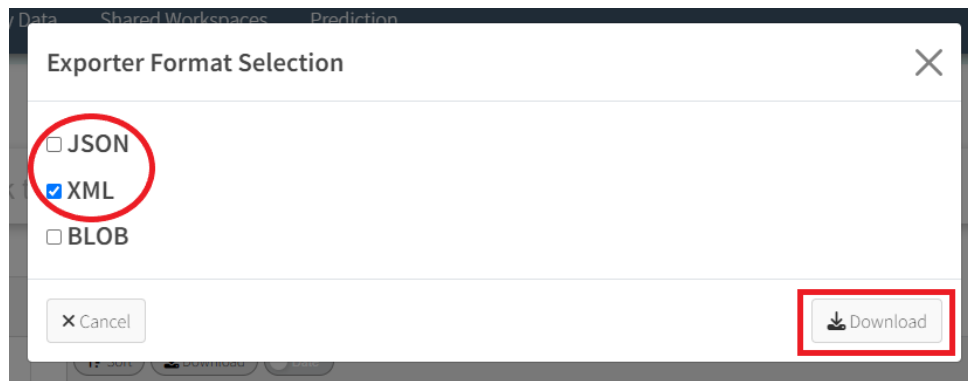
Filter by Test Select All
☐ Rutting
☐ Marshall
☐ ITS

From AsphaltMine 25
tf Sort Download Date

☐	ITS (Version 1)	
☐	Rutting (Version 1)	
☒	AC8_ORGUNKNOWN_2020_AMUID25 ITS (Version 1)	
☐	ITS (Version 1)	
☒	ITS (Version 1)	
☐	Rutting (Version 1)	
☐	ITS (Version 1)	
☒	Marshall (Version 1)	
☒	Marshall (Version 1)	
☐	Rutting (Version 1)	

« 1 2 3 »

4) In the popup message, select either XML or JSON, or both, then click "Download".



The files will be downloaded and organized into a ZIP folder.

Delete Records

Overview

Users can delete a record through either the 'My Data' page or the 'Shared Workspaces' page. Deleting a record is permanent and can only be performed by the record's owner.

Delete using "My Data" page

This method allows users to delete a record through the "My Data" page. To do so, follow these steps:

- 1) Navigate to the "My Data" page.

Total number of records: 3

Name	Test	Last Modification date	Workspace	Actions
AC8_ORGUNKNOWN_2020_AMUID25	Rutting (Version 1)	Aug 10, 2020 10:00:00	My Data	
AC8_ORGUNKNOWN_2020_AMUID25	ITS (Version 1)	Aug 10, 2020 10:00:00	Global Public Workspace	
AC8_ORGUNKNOWN_2020_AMUID25	Marshall (Version 1)	Aug 10, 2020 10:00:00	Global Public Workspace	

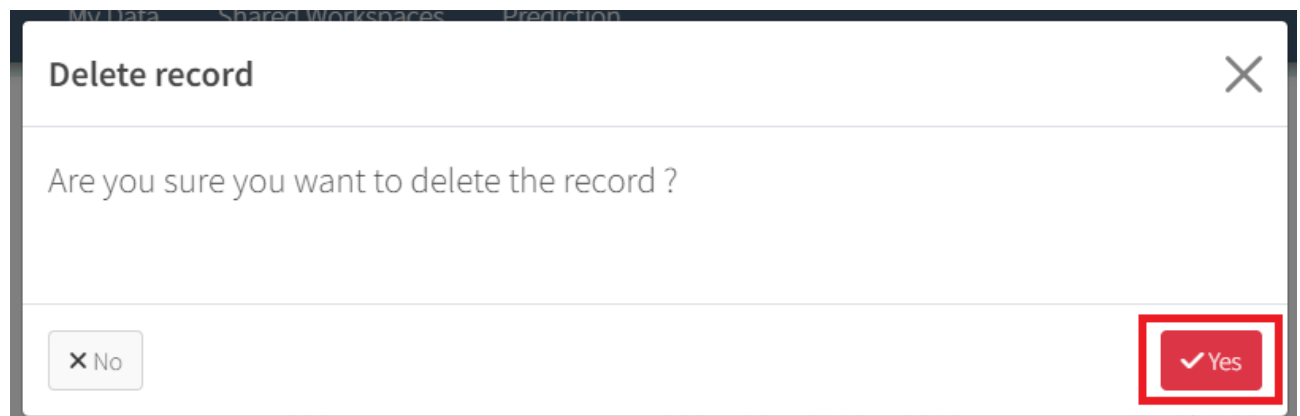
- 2) Identify the record to be deleted, click its Action button in the last column, and then click 'Delete'.

Total number of records: 3

Name	Test	Last Modification date	Workspace	Actions
AC8_ORGUNKNOWN_2020_AMUID25	Rutting (Version 1)	Aug 10, 2020 10:00:00	My Data	
AC8_ORGUNKNOWN_2020_AMUID25	ITS (Version 1)	Aug 10, 2020 10:00:00	Global Public Workspace	 
AC8_ORGUNKNOWN_2020_AMUID25	Marshall (Version 1)	Aug 10, 2020 10:00:00	Global Public Workspace	

 View
 Edit
 Change workspace
 Change Owner
 Delete

- 3) In the popup message, click "Yes" to confirm.



The file will be permanently deleted from the database and from its workspace, if applicable.

Delete using "Shared Workspaces" page

For this method, the record must be in one of the workspaces.

This method allows users to delete a record through the "Shared Workspaces" page. To do so, follow these steps:

1) Navigate to the "Shared Workspaces" page.

Total number of workspaces: 3

Create workspace

Title	Owner	Can read	Can write	Public	Actions
Global Public Workspace	GLOBAL	True	False	True	View Content
Test	Test	True	False	False	View Content
Test	Test	True	True	False	View Content Manage Access Set public Delete

2) Choose the target workspace by clicking on the "View Content" Button.

Total number of workspaces: 3

Create workspace

Title	Owner	Can read	Can write	Public	Actions
Global Public Workspace	GLOBAL	True	False	True	View Content
Test	Test	True	False	False	View Content
Test	Test	True	True	False	View Content Manage Access Set public Delete

3) Within the selected workspace, identify the record to be deleted, click its Action button in the last column, and then click 'Delete'.

Global Public Workspace

Data (8)

Files (0)

Previous page

Name	Test	Last Modification date	User	Actions
AC8_ORGUNKNOWN_2020_AMUID25	Rutting (Version 1)	2020-10-20 10:00:00	Test	
AC8_ORGUNKNOWN_2020_AMUID25	ITS (Version 1)	2020-10-20 10:00:00	Test	
AC8_ORGUNKNOWN_2020_AMUID25	Rutting (Version 1)	2020-10-20 10:00:00	Test	View
AC8_ORGUNKNOWN_2020_AMUID25	Marshall (Version 1)	2020-10-20 10:00:00	Test	Edit
AC8_ORGUNKNOWN_2020_AMUID25	Rutting (Version 1)	2020-10-20 10:00:00	Test	Change workspace
AC8_ORGUNKNOWN_2020_AMUID25	Marshall (Version 1)	2020-10-20 10:00:00	Test	Change Owner
AC8_ORGUNKNOWN_2020_AMUID25	Marshall (Version 1)	2020-10-20 10:00:00	Test	Delete
AC8_ORGUNKNOWN_2020_AMUID25	ITS (Version 1)	2020-10-20 10:00:00	Test	

4) In the popup message, click "Yes" to confirm.

Delete record

Are you sure you want to delete the record ?

No

Yes

The file will be permanently deleted from both the workspace and the database.

Other Functionalities

Change Record's Workspace

Overview

A workspace is a collection of records within a database. Each workspace has an owner who controls access by assigning permissions for who can read from it and who can add records to it. Once a record is added to a workspace, the workspace's rules determine its visibility and accessibility. In other words, users authorized by the workspace owner to view records in that workspace will also be able to view this record, and the same applies to users authorized to write to the workspace.

Initially, when a record is created, it is not assigned to any workspace and is only visible to its owner. In this case, the record's workspace in the 'My Data' page will be listed as 'None', as shown below.

Name	Test	Last Modification date	Workspace	Actions
AC8_ORGUNKNOWN_2020_AMUID25	Rutting (Version 1)	Aug 11, 2020 12:00:00	Global Public Workspace	
AC8_ORGUNKNOWN_2020_AMUID25	Marshall (Version 1)	Aug 11, 2020 12:00:00	None	
AC8_ORGUNKNOWN_2020_AMUID25	Rutting (Version 1)	Aug 11, 2020 12:00:00	None	
AC8_ORGUNKNOWN_2020_AMUID25	ITS (Version 1)	Aug 11, 2020 12:00:00	None	
AC8_ORGUNKNOWN_2020_AMUID25	Rutting (Version 1)	Aug 11, 2020 12:00:00	Global Public Workspace	
AC8_ORGUNKNOWN_2020_AMUID25	Rutting (Version 1)	Aug 11, 2020 12:00:00	None	
AC8_ORGUNKNOWN_2020_AMUID25	Rutting (Version 1)	Aug 11, 2020 12:00:00	None	

The owner of the record can change its workspace from 'None' to another workspace, and then back to 'None' whenever desired.

To change or assign a record to a workspace, the owner can do so via either the 'My Data' page or the 'Shared Workspaces' page.

Change Workspace via "My Data" page

This method allows users to change the workspace of a record via the "My Data" page. It can only be used if the record is part of a workspace. To change the workspace of the record, follow these steps:

- 1) Navigate to the "My Data" page.

Total number of records: 3

Name	Test	Last Modification date	Workspace	Actions
AC8_ORGUNKNOWN_2020_AMUID25	Rutting (Version 1)	Aug 11, 2020 12:00:00	None	
AC8_ORGUNKNOWN_2020_AMUID25	ITS (Version 1)	Aug 11, 2020 12:00:00	Global Public Workspace	
AC8_ORGUNKNOWN_2020_AMUID25	Marshall (Version 1)	Aug 11, 2020 12:00:00	Global Public Workspace	

- 2) Identify the record for which the workspace needs to be changed, click the Action button in the last column, and then click 'Change workspace'.

Total number of records: 3

Name	Test	Last Modification date	Workspace	Actions
AC8_ORGUNKNOWN_2020_AMUID25	Rutting (Version 1)	Aug 10, 2020 10:00:00	Global Public Workspace	
AC8_ORGUNKNOWN_2020_AMUID25	ITS (Version 1)	Aug 10, 2020 10:00:00	Global Public Workspace	
AC8_ORGUNKNOWN_2020_AMUID25	Marshall (Version 1)	Aug 10, 2020 10:00:00	Global Public Workspace	

 View

 Edit

 Change workspace

 Change Owner

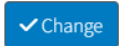
 Delete

3) In the popup window, select the new workspace and then click 'Change'.

Change workspace

Select a new workspace to assign the record:

 Cancel

 Change

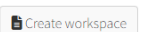
Change Workspace via "Shared Workspaces" page

For this method, the record must be in one of the workspaces.

This method allows users to change ownership of a record via the "Shared Workspaces" page. To do so, follow these steps:

1) Navigate to the "Shared Workspaces" page.

Total number of workspaces: 3



Title	Owner	Can read	Can write	Public	Actions
Global Public Workspace	GLOBAL	True	False	True	
Test	Test	True	False	False	
Test	Test	True	True	False	   

2) Choose the target workspace by clicking on the "View Content" Button.

Total number of workspaces: 3



Title	Owner	Can read	Can write	Public	Actions
Global Public Workspace	GLOBAL	True	False	True	
Test	Test	True	False	False	
Test	Test	True	True	False	   

- 4) Within the selected workspace, identify the record for which the workspace needs to be changed, click the Action button in the last column, and then click 'Change workspace'.

Name	Test	Last Modification date	User	Actions
AC8_ORGUNKNOWN_2020_AMUID25	Rutting (Version 1)	2020-10-20 10:10:10	user	
AC8_ORGUNKNOWN_2020_AMUID25	Rutting (Version 1)	2020-10-20 10:10:10	user	
AC8_ORGUNKNOWN_2020_AMUID25	ITS (Version 1)	2020-10-20 10:10:10	user	

 View
 Edit
 Change workspace
 Change Owner
 Delete

- 5) In the popup window, select the new workspace and then click 'Change'.

Change workspace

Select a new workspace to assign the record:

✕ Cancel

✓ Change

Change Record's Owner

Overview

Each record is initially owned by the user who added it to AsphaltMine. The owner has the ability to transfer ownership to another user at any stage. Once ownership is transferred, the previous owner will no longer have the rights to delete or change the ownership of the record. Only the new owner will have these rights.

Ownership can be transferred via the 'My Data' page or the 'Shared Workspaces' page.

Change Owner via "My Data" page

This method allows users to change ownership of a record via the "My Data" page. To do so, follow these steps:

- 6) Navigate to the "My Data" page.

Total number of records: 3

Name	Test	Last Modification date	Workspace	Actions
AC8_ORGUNKNOWN_2020_AMUID25	Rutting (Version 1)	Aug 10, 2020 10:00:00	Public	
AC8_ORGUNKNOWN_2020_AMUID25	ITS (Version 1)	Aug 10, 2020 10:00:00	Global Public Workspace	
AC8_ORGUNKNOWN_2020_AMUID25	Marshall (Version 1)	Aug 10, 2020 10:00:00	Global Public Workspace	

- 7) Identify the record for which ownership needs to be changed, click the Action button in the last column, and then click 'Change Owner'.

Total number of records: 3

Name	Test	Last Modification date	Workspace	Actions
AC8_ORGUNKNOWN_2020_AMUID25	Rutting (Version 1)	Aug 10, 2020 10:00:00	Public	
AC8_ORGUNKNOWN_2020_AMUID25	ITS (Version 1)	Aug 10, 2020 10:00:00	Global Public Workspace	
AC8_ORGUNKNOWN_2020_AMUID25	Marshall (Version 1)	Aug 10, 2020 10:00:00	Global Public Workspace	

View

Edit

Change workspace

Change Owner

Delete

- 8) In the popup window, select the new owner and then click 'Change'.

Change owner

The owner of the document(s) will be modified.

Please select the new owner

Cancel

Change

This action will permanently change the ownership of the record and transfer ownership rights to the selected user.

Change Owner via "Shared Workspaces" page

For this method, the record must be in one of the workspaces.

This method allows users to change ownership of a record via the "Shared Workspaces" page. To do so, follow these steps:

3) Navigate to the "Shared Workspaces" page.

Total number of workspaces: 3 Create workspace

Title	Owner	Can read	Can write	Public	Actions
Global Public Workspace	GLOBAL	True	False	True	View Content
Page	User/Profile	True	False	False	View Content
Page	User/Profile	True	True	False	View Content Manage Access Set public Delete

4) Choose the target workspace by clicking on the "View Content" Button.

Total number of workspaces: 3 Create workspace

Title	Owner	Can read	Can write	Public	Actions
Global Public Workspace	GLOBAL	True	False	True	View Content
Page	User/Profile	True	False	False	View Content
Page	User/Profile	True	True	False	View Content Manage Access Set public Delete

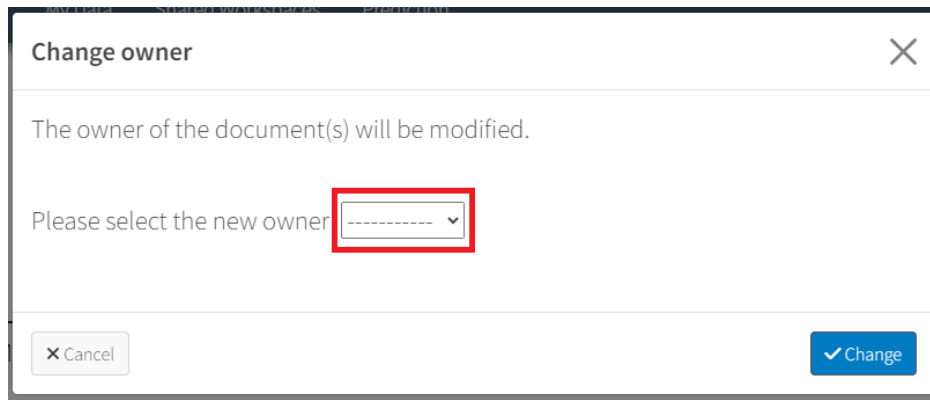
5) Within the selected workspace, identify the record for which ownership needs to be changed, click the Action button in the last column, and then click 'Change Owner'.

Global Public Workspace

Data (8) Files (0) Previous page

Name	Test	Last Modification date	User	Actions
AC8_ORGUNKNOWN_2020_AMUID25	Rutting (Version 1)	2020-11-20 10:00:00	User/Profile	⋮
AC8_ORGUNKNOWN_2020_AMUID25	ITS (Version 1)	2020-11-20 10:00:00	User/Profile	⋮
AC8_Page 2020 AMUID25	Rutting (Version 1)	2020-11-20 10:00:00	User/Profile	View
AC8_Testing 2020 AMUID25	Marshall (Version 1)	2020-11-20 10:00:00	User/Profile	Edit
AC8_ORGUNKNOWN_2020_AMUID25	Rutting (Version 1)	2020-11-20 10:00:00	User/Profile	Change workspace
AC8_ORGUNKNOWN_2020_AMUID25	Marshall (Version 1)	2020-11-20 10:00:00	User/Profile	Change Owner
AC8_ORGUNKNOWN_2020_AMUID25	Marshall (Version 1)	2020-11-20 10:00:00	User/Profile	Delete
AC8_ORGUNKNOWN_2020_AMUID25	ITS (Version 1)	2020-11-20 10:00:00	User/Profile	⋮

6) In the popup window, select the new owner and then click 'Change'.

A dialog box titled "Change owner" with a close button (X) in the top right corner. The text inside reads: "The owner of the document(s) will be modified." followed by "Please select the new owner". Below this text is a dropdown menu with a dashed line and a downward arrow, which is highlighted by a red rectangular box. At the bottom left is a "Cancel" button with an 'X' icon, and at the bottom right is a "Change" button with a checkmark icon.

Change owner

The owner of the document(s) will be modified.

Please select the new owner

----- ▾

✕ Cancel

✓ Change

This action will permanently change the ownership of the record and transfer ownership rights to the selected user.